

```
Import java.util.ArrayList;
```

```
Import java.util.Scanner;
```

```
Public class CityArrayList {
```

```
    Public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        ArrayList<String> cities = new ArrayList<>();
```

```
        System.out.print("  Enternumber of cities:"  );
```

```
        Int n = sc.nextInt();
```

```
        Sc.nextLine();
```

```
        System.out.println("  Enter city names:"  );
```

```
        For (int l = 0; l < n; l++) {
```

```
            Cities.add(sc.nextLine());
```

```
        }
```

```
        While(true) {
```

```
            System.out.println("  \n===== MENU ====="  );
```

```
            System.out.println("  1. Append a City"  );
```

```
            System.out.println("  2. Insert a City at Index"  );
```

```
            System.out.println("  3. Search a City"  );
```

```
            System.out.println("  4. Display Cities Starting with a Letter"  );
```

```
System.out.println(" 5.Exit" );
```

```
System.out.print(" Enteryourchoice:" );
```

```
Int choice = sc.nextInt();
```

```
Sc.nextLine();
```

```
Switch(choice){
```

```
Case 1:
```

```
System.out.print(" Enter city to append:" );
```

```
String city = sc.nextLine();
```

```
Cities.add(city);
```

```
System.out.println(" Updated List:" + cities);
```

```
Break;
```

```
Case 2:
```

```
System.out.print(" Enter index:" );
```

```
Int index = sc.nextInt();
```

```
Sc.nextLine();
```

```
System.out.print(" Enter city to insert:" );
```

```
City = sc.nextLine();
```

```
If (index >= 0 && index <= cities.size()) {
```

```
Cities.add(index, city);
```

```
        System.out.println(" Updated List:" + cities);  
    }else{  
        System.out.println(" Invalid Index" );  
    }  
    Break;
```

Case 3:

```
    System.out.print(" Enter city to search:" );  
    City = sc.nextLine();  
  
    Int pos = cities.indexOf(city);  
    If (pos != -1){  
        System.out.println(" City Found at Index:" + pos);  
    }else{  
        System.out.println(" City Not Found" );  
    }  
    Break;
```

Case 4:

```
    System.out.print(" Enter starting letter:" );  
    Char ch = sc.next().charAt(0);  
  
    System.out.print(" Cities Starting with ' " + ch + " ' :" );  
    Boolean found = false;
```

```
For(String c: cities){  
    If (Character.toUpperCase(c.charAt(0)) == Character.toUpperCase(ch)){  
        System.out.print(c + " " );  
        Found = true;  
    }  
}
```

```
    If(!found){  
        System.out.print(" No cities found" );  
    }  
    System.out.println();  
    Break;
```

Case 5:

```
    System.out.println(" Program Ended" );  
    Sc.close();  
    System.exit(0);
```

Default:

```
    System.out.println(" Invalid Choice" );  
}  
  
}  
  
}
```

}